Noetic Learning Math Contest 2026 Spring - Grade 6

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1) $2026 \times \frac{1}{2} = ?$

Correct! Answer: 1013

2) A cashier has an unlimited supply of pennies (1¢), nickels (5¢), dimes (10¢), and quarters (25¢). What is the minimum number of coins needed to make exactly 43¢?

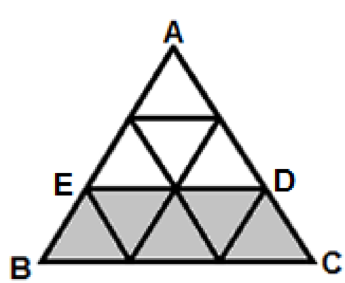
✗ Answer: 7 coins

Correct Answer:
6

3) A marching band practices its formation. At first, the entire band forms a rectangle that has 7 rows with the same number of people in each row. Then the entire band forms another rectangle that has 6 rows with the same number of people in each row. The number of people in the band is between 150 and 200. How many people are in the band?

Correct! Answer: 168 people

4) Triangle ABC is made of smaller identical triangles, as shown below. The area of triangle ABC is 72 square inches. What is the area of the shaded trapezoid BCDE?



Correct! Answer: 40 square inches

5) Rick burns 60 calories during a 12-minute walk. If he keeps burning calories at this rate, how many minutes in total will it take him to burn 540 calories?

Correct! Answer: 108 minutes

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6) For every 3 tulip bulbs Cynthia planted, she planted 5 iris bulbs. She planted 12 more iris bulbs than tulip bulbs. How many tulip bulbs did she plant in all?

Correct! Answer: 18 tulip bulbs

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7) Mom's age is $\frac{3}{4}$ of Dad's age, and Dad's age is $\frac{2}{3}$ of Grandma's age. Mom is 36 years old. How old is Grandma?

Correct! Answer: 72 years old

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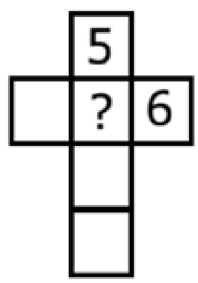
8) In the figure below, place the numbers 1, 2, 3, and 4 in the empty boxes, using each number exactly once. The sum of the three numbers in the horizontal row must be 11, and the sum of the four numbers in the vertical column must also be 11. What number should go in the box with the question mark?



Correct! Answer: 1

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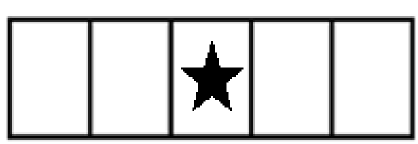
9) In the figure below, how many rectangles of any size contain the star?



X Answer: 5 rectangles

Correct Answer:
9

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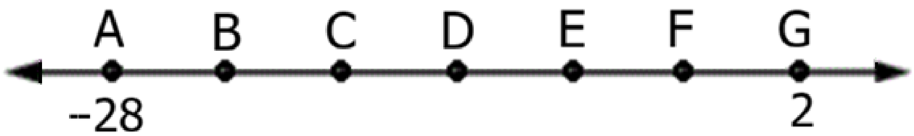
10) On the number line below, points A through G are equally spaced. Point A represents −28 and point G represents 2. What number does point E represent?



Correct! Answer: -8

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11) The four-digit number 2178 has digits 1, 2, 7, and 8. What other positive four-digit number is a multiple of 2178 and uses each of these digits exactly once?

Correct! Answer: 8712

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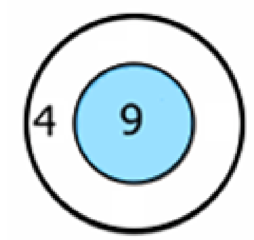
12) Payton played a game of darts using the dart board shown below. Each of his darts scored either 4 points or 9 points. He scored a total of 51 points. How many of his darts scored 9 points?



Correct! Answer: 3 darts

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13) During the first half of the semester, Judy had four math tests with an average score of 83. During the second half of the semester, she had six math tests with an average score of 93. What was her average score on all ten tests?

Correct! Answer: 89

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14) Isaac collects toy cars. If he puts all his toy cars in groups of 2, he will have 1 car left over. If he puts them in groups of 3, he will have 2 cars left over. If he puts them in groups of 5, he will have 4 cars left over. Isaac has fewer than 50 toy cars. How many toy cars does Isaac have?

Correct! Answer: 29 toy cars

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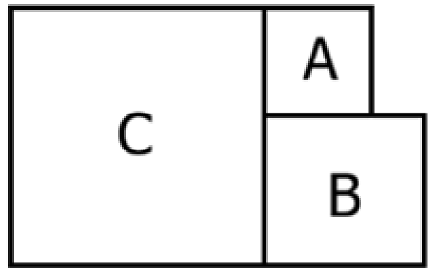
15) Garvin has three square tiles: A, B, and C. He glued them together to form a new shape, as shown below. The perimeter of tile A is 16 inches. The perimeter of tile B is 24 inches. What is the perimeter of the new shape?



❌ Answer: 40 inches

Correct Answer:
52

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16) In the addition problem below, the same letters represent the same digit, and different letters represent different digits. What digit does the letter A represent?

$$
\begin{array}{r}
A\ A \\
A\ B \\
+\ A\ B \\
\hline
C\ C\ A
\end{array}
$$

Correct! Answer: 7

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17) There are 40 guests at a party, and they are arranged into 20 pairs. Each pair shares either one pecan pie or one cherry pie. A pair is served pecan pie if neither person is allergic to nuts, and a pair is served cherry pie if at least one person is allergic to nuts. Of all the guests, 14 are allergic to nuts, and 8 pairs are served cherry pie. In how many pairs are both people allergic to nuts?

Correct! Answer: 6 pairs

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18) It takes Ryan 6 hours to mow a soccer field by himself. It takes Seth 12 hours to mow the same soccer field by himself. Ryan starts mowing the field alone. Then, 3 hours later, Seth joins him. How many more hours will it take them to finish mowing the field together? Note: Ryan and Seth each mow at a constant rate.

Correct! Answer: 2 more hour(s)

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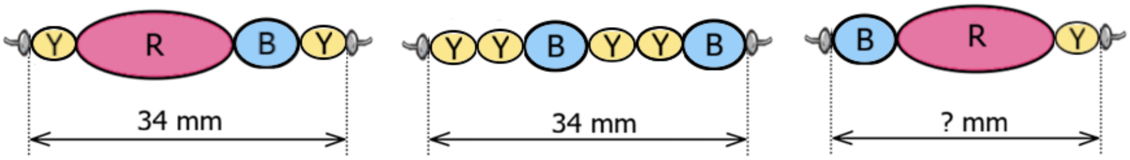
19) Georgia makes strings of beads using three different colored beads: blue (B), red (R), and yellow (Y). All beads of the same color have the same length, and a blue bead is 2 mm longer than a yellow bead. She first makes two strings that are each 34 mm long, as shown below. Then, she makes a third string using 1 blue bead, 1 red bead, and 1 yellow bead. How long is the third string?



Correct! Answer: 29 mm

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20) Today, $\frac{3}{4}$ of the students at Washington Elementary checked out exactly 1 book each from the school library. The remaining 105 students did not check out any books. The books that were checked out today represent exactly 5% of the total number of books in the library's collection. What is the total number of books in the library's collection?

Correct! Answer: 6300 books

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